PROJECT AND TEAM INFORMATION

Project Title

FUND NEXUS – FINTECH/FUND MANAGEMENT SYSTEM

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Crowdfunding has become a widely used way for individuals and communities to raise money for education, medical emergencies, marriages, disaster relief and community development. However, most existing platforms operate largely on trust: campaigns go live almost immediately, fund utilisation is rarely tracked after collection, and duplicate or exaggerated campaigns for the same event are common, especially during disasters. These gaps allow fake campaigns, diversion of funds to unintended purposes, and donor fatigue caused by dozens of near-identical campaigns for one incident. This is important because crowdfunding directly affects vulnerable people who genuinely need help — a single high-profile fraud case damages trust in the entire ecosystem and discourages future donors from contributing to legitimate causes. FundNexus is motivated by the need for a platform that does not merely collect donations, but actively verifies fundraisers, screens campaigns using rule-based risk analysis, groups disaster-related campaigns intelligently, and tracks how collected funds are actually spent — bringing accountability into a process that today relies almost entirely on goodwill.

State of the Art / Current solution

Today, most crowdfunding is handled by large commercial platforms (e.g. GoFundMe, Ketto, Milaap) that primarily focus on payment collection and campaign discovery. Verification, where it exists, is often limited to basic document uploads with manual, opaque review. These platforms rarely expose donors to post-collection fund utilisation, do not systematically detect duplicate campaigns arising from the same disaster, and have no visible mechanism to flag overfunding when many campaigns target the same need. Risk assessment, when present, is a black-box internal process. There is no widely available academic-grade, transparent, rule-based system that a student team can study, extend or replicate to understand how trust and safety mechanisms in crowdfunding could realistically be engineered.

Project Goals and Milestones

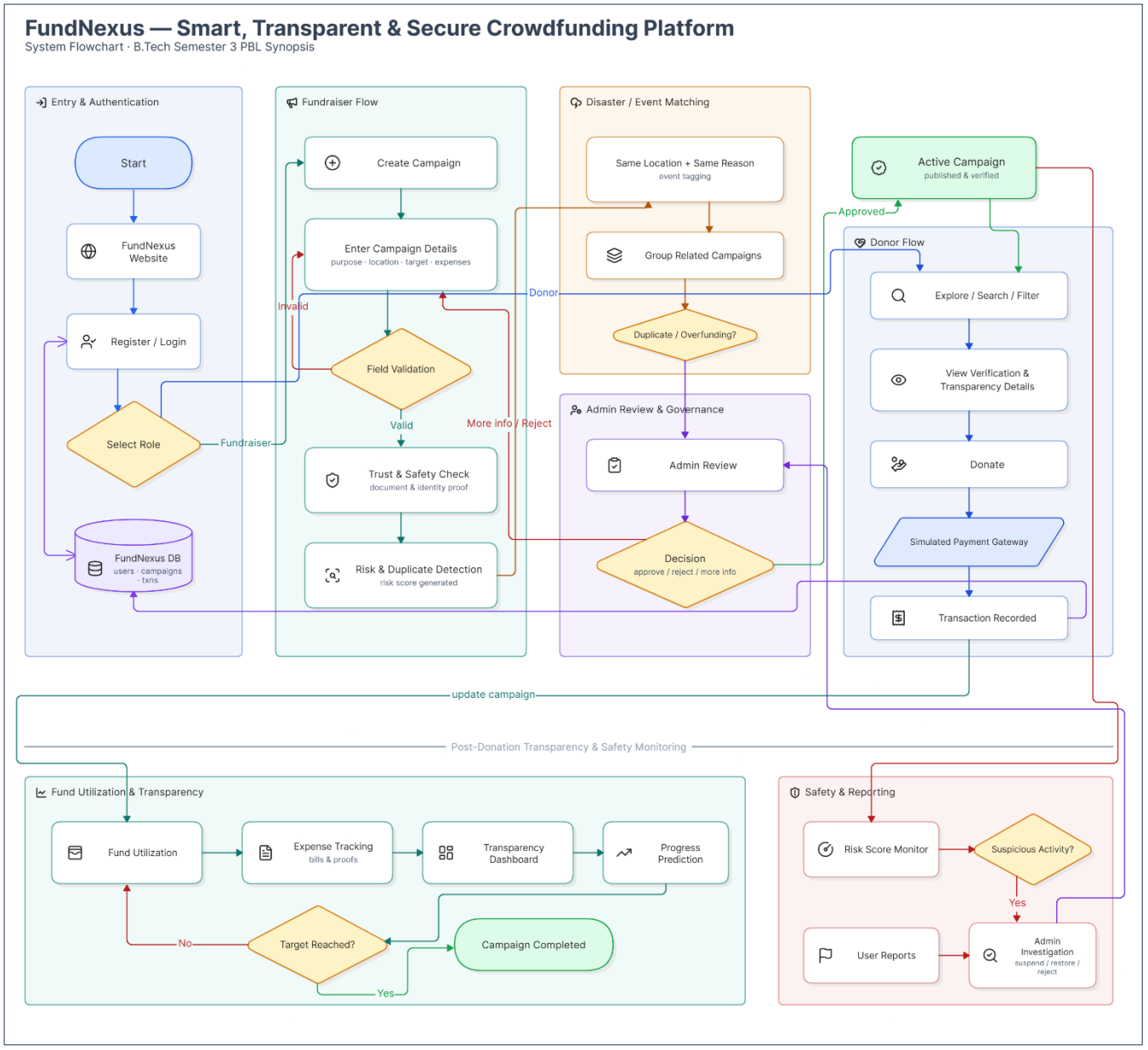
The overall goal is to design and implement a working prototype crowdfunding platform that demonstrates verification, transparency and rule-based trust & safety mechanisms using core Semester 3 CSE concepts (OOP, data structures, algorithms, web development, and file/database handling).

- Milestone 1 – Requirement analysis, module design, database schema and system architecture finalisation.
- Milestone 2 – User management module: registration, login, role-based access (Donor / Fundraiser / Admin).
- Milestone 3 – Campaign management module: creation, lifecycle states, category/location tagging.
- Milestone 4 – C/C++ computation engine: risk scoring, similarity detection, ranking and prediction algorithms.
- Milestone 5 – Disaster/Event grouping module and overfunding detection logic.
- Milestone 6 – Fund utilisation tracking, transparency dashboard and simulated donation system.
- Milestone 7 – Reporting workflow, admin dashboard, integration testing and final documentation.

Project Approach

FundNexus will be built as a layered web application. The frontend (HTML, CSS, JavaScript) will provide role-specific interfaces for Donors, Fundraisers and Administrators. A backend server will manage authentication, campaign lifecycle, and communication between the web layer, the database, and a dedicated C/C++ computation engine. The C/C++ modules will implement the computationally-oriented logic explicitly required for this PBL — rule-based risk scoring, campaign similarity detection, weighted campaign ranking, overfunding calculation and progress prediction
— using arrays/vectors, linked lists, stacks, queues, and standard searching and sorting algorithms, invoked by the backend via file exchange or function calls. Campaign, user and donation records will be stored in a relational database (or structured files, if simplified). Donations will be simulated (no real payment gateway), generating a mock transaction ID and receipt. This approach keeps the architecture realistic for a team while still giving every core DSA and OOP concept a genuine, non-trivial role in the system rather than treating them as an afterthought.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The project will deliver: (1) a functional web-based crowdfunding platform with Donor, Fundraiser and Administrator roles; (2) a campaign verification workflow with a mandatory admin review queue; (3) a C/C++ computation engine implementing risk scoring, similarity/duplicate detection, campaign ranking and completion prediction; (4) a disaster/event grouping module with combined overfunding analysis; (5) a fund-utilisation transparency dashboard comparing planned versus reported expenses; (6) a simulated donation and receipt-generation system; (7) an administrative dashboard with platform-wide statistics and a reporting/complaints workflow; and (8) complete project documentation including this proposal, the design report, source code, and a demonstration/viva-ready presentation.

Assumptions

Payment processing is simulated only; no real bank, UPI or card transactions occur. Identity/document verification is a prototype-level check (format and completeness validation), not real government-level KYC. Risk scoring, similarity detection and ranking are deterministic rule-based algorithms, not machine learning. The system is evaluated with a moderate, simulated dataset of users and campaigns rather than production-scale traffic

References

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to algorithms* (4th ed.). MIT Press.
2. Deitel, P., & Deitel, H. (2020). *C++: How to program* (11th ed.). Pearson. GoFundMe. (n.d.). *GoFundMe: Online fundraising platform*.
3. Ketto. (n.d.). *Ketto: Crowdfunding platform*.
4. Milaap. (n.d.). *Milaap: Crowdfunding platform*.
5. Mozilla. (n.d.). *MDN Web Docs*.
6. W3Schools. (n.d.). *W3Schools online web tutorials*.